TWO EQUIVALENT FORMULATIONS OF THE FAILURE OF FLATNESS IN DIMENSIONS BELOW N

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Transcendence degree as the dimension of the closure of a point in the fibre, and the dimension formula for fibres after base change. Varieties, label varieties-lemma-dimension-locally-algebraic (proof lines 6290 and 6293); Morphisms, labels morphisms-lemma-base-change-module-flat and morphisms-lemma-dimension-fibre-after-base-change (proof lines 6296-6297). The first converts between the dimension of a closed subset of the fibre and the transcendence degree at its points, giving the equivalence even fibre by fibre, and the second two give the base change statement.
- Openness of the flat locus inside a fibre. Same chapter flat.tex, label lemma-open-in-fibre-where-flat, invoked in the statement at proof line 6278. Makes the set of points of a fibre where the module is not flat a closed subset, so that its dimension is defined.

Source reference. Original lemma 05M9: sources/stacks-flat/pinned-original-source.tex, source lines 2676–2676. Navigation pointer for \ref{lemma-open-in-fibre-where-flat} used inside the counted span (line 6278); the argument that consumes it is retained in the span.

Source reference. Original lemma 01U8: sources/stacks-chow/tags.txt, source lines 2401–2401. Navigation pointer for \ref{morphisms-lemma-base-change-module-flat} used inside the counted span (line 6296); the argument that consumes it is retained in the span.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 02FY: sources/stacks-chow/tags.txt, source lines 3209–3209. Navigation pointer for \ref{morphisms-lemma-dimension-fibre-after-base-change} used inside the counted span (line 6297); the argument that consumes it is retained in the span.

Source reference. Original lemma 0A21: sources/stacks-chow/tags.txt, source lines 12689–12689. Navigation pointer for \ref{varieties-lemma-dimension-locally-algebraic} used inside the counted span (line 6290); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module of finite type. Let $n \geq 0$. The following are equivalent*

(1) *for $s \in S$ the closed subset $Z \subset X_s$ of points where $\mathcal{F}$ is not flat over S (see Lemma 05M9) satisfies $\dim(Z) < n$, and*
(2) *for $x \in X$ such that $\mathcal{F}$ is not flat at x over S we have $trdeg_{\kappa(f(x))}(\kappa(x)) < n$.*

If this is true, then it remains true after any base change.

Proof. Let $x \in X$ be a point over $s \in S$. Then the dimension of the closure of $\{x\}$ in X_s is $\mathrm{trdeg}_{\kappa(s)}(\kappa(x))$ by Varieties, Lemma 0A21. Conversely, if $Z \subset X_s$ is a closed subset of dimension d, then there exists a point $x \in Z$ with $\mathrm{trdeg}_{\kappa(s)}(\kappa(x)) = d$ (same reference). Therefore the equivalence of (1) and (2) holds (even fibre by fibre). The statement on base change follows from Morphisms, Lemmas 01U8 and 02FY. □